Total No. of Pages: 2 Roll No.....

VITH SEMESTER B.Tech.(Software Engineering)

MID SEMESTER EXAMINATION (March, 2019)

CO_304 ARTIFICIAL INTELLIGENC

Paper Code

Time: 1:30 Hours Max. Marks: 25

Note:	Answer all questions. Assume suitable missing data, if any.

Q1. Answer to the point for any five from following:

(i) Define the production system using water jug problem.

(ii) Analyze the problem characteristics of Travelling sales man problem.

(iii). Why is A* search superior to best first search?

(iv) What do you understand by inferential efficiency in Knowledge representation method.

(v) Define alpha-beta cut- off and their utility.

(vi) Describe the control strategy of constraint satisfaction problem

[5x3]

Q2. Consider a state space where start space is number 1 and successor for state numbered n is 3n and 3n-1. Draw the search tree. List the order in which nodes will be visited using following search techniques if goal node is numbered as 24

(i) Depth first search, (ii) Iterative deepening search.

[2]

Q3. Explain the traversing of graph in fig 1 using AO*procedure [2].

Q4. Work out first two steps of A* algorithm to solve Eight puzzle problem where initial state and goal state are described below:

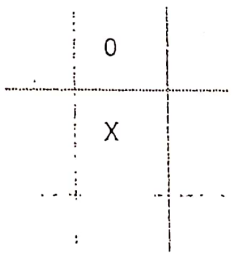
5	3	6
7	.	2
4	1	8

3	7	6
5	1	2
4		8

[3]

Q5. Illustrate the steps of Minimax procedure to make a move for Maximizing player in response to following board position.

[3]



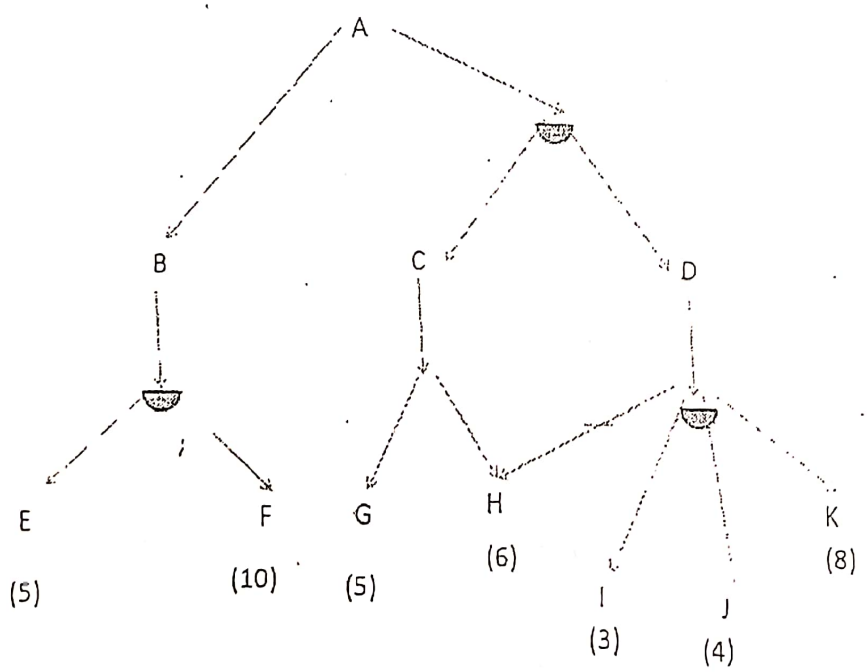


Fig.1